PRE-IDEATHONTEMPLATE

GENESIS IDEATHON

TEAM &DOMAIN

Introduce your team and the domain you have chosen

Team Name – LOGIC LOOM Team Members :

- Badhma Sree Vignesh S (CB.EN.U4CCE25009)
- Darshan B (CB.EN.U4CCE25015)
- Prithivram B (CB.EN.U4CCE25040)
- Mahathi S (CB.EN.U4CCE25028)

Selected Domain – Domian (5) - Food Security, Agriculture and Rural Transformation

Problem Statement – AI - Powered Food Surplus Prediction, Prevention and Redistribution System

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DOMAIN PROBLEM IN OUR WORDS

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What exactly needs to be solved?

1.Large quantities of perfectly edible food are wasted by restaurants, hotels, supermarkets, hostels, cafeterias and events because actual demand is difficult to predict.

2.Businesses often realize they have more food than needed only after it has already been prepared, making prevention difficult.

3.When surplus occurs, finding a suitable nearby recipient and arranging pickup within the limited time available is challenging.

4.Most existing approaches focus on redistributing food after surplus occurs, rather than preventing unnecessary overproduction.

Why is it important right now?

1.Food waste means losing money, food, water, energy and human effort used to produce it.

2.Rising food costs make unnecessary overproduction a direct financial loss for businesses.

3.At the same time, many communities and organizations face food availability challenges.

4.Wasted food creates avoidable environmental impact through wasted resources and disposal.

5.The challenge is to predict excess early, prevent unnecessary production, and quickly redirect unavoidable surplus.

We don’t just need to find someone to take leftover food. We need to predict excess before it happens, prevent unnecessary production, and get unavoidable surplus to someone who can use it before it becomes waste.

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WHO IS AFFECTED?

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Target users, stakeholders, beneficiaries

Primary Users

- Restaurants, hotels, supermarkets, hostels, cafeterias & event organizers — generate surplus food.
- NGOs, food banks & shelters — receive and redistribute usable food.

Secondary Stakeholders

- Logistics partners — collect and deliver surplus on time.
- Food-safety authorities — ensure safe handling and redistribution.
- Local communities — benefit from better food access.

Who Benefits?

- Businesses: Lower food waste & costs
- Communities: Better access to available food
- Environment: Less resource waste & disposal
- Society: Better food utilization & reduced food insecurity

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CURRENT SITUATION & CONTEXT

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Existing scenario, background and how things work today

Current Situation

- Businesses estimate food demand using past sales, bookings and experience.
- Unexpected changes in demand lead to excess food and unnecessary waste.
- Surplus is often identified only after food has been prepared.

Existing Solutions

Food-sharing platforms sell surplus food at discounted prices.
NGOs and food recovery networks collect and redistribute excess food.

The Existing Gap

- Demand prediction and surplus redistribution are often handled separately.
- Businesses need a more coordinated way to predict surplus, prevent overproduction and redistribute unavoidable excess quickly.

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KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

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1. Unpredictable Demand:

Root Cause: Customer demand changes with weekends, holidays, events, weather and local patterns.

Consequence: Businesses often prepare more food than actually needed.

2. Surplus Detected Too Late:

Root Cause: Excess is usually recognized only after food has already been prepared.

Consequence: The opportunity to prevent unnecessary overproduction is already lost.

3. Difficult Real-Time Matching:

Root Cause: Finding a nearby recipient for the available food and quantity is difficult.

Consequence: Valuable time is lost searching for suitable recipients.

4. Time-Critical Pickup:

Root Cause: Perishable food has a limited usable window and requires timely collection.

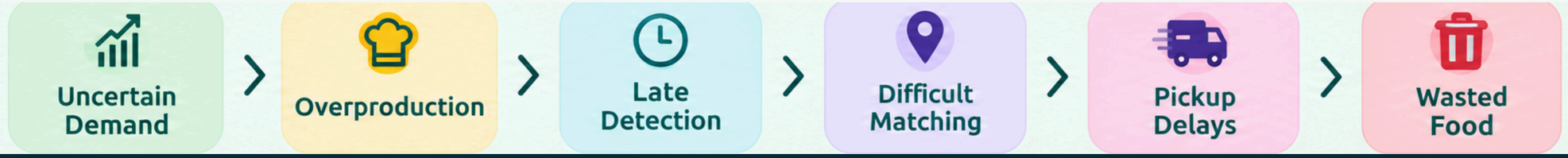
Consequence: Even suitable food can become unusable before pickup

5. Repeated Waste:

Root Cause: Businesses often lack a system that learns from previous surplus patterns.

Consequence: The same overproduction mistakes can happen again and again.

THE CHAIN OF FAILURE



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EXISTING SOLUTIONS / APPROACHES

What exists today-and the gap that remains

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1.FOOD DONATION NETWORKS:

Examples: Robin Hood Army, Feeding India

What they do: Connect surplus food with charities and communities in need.

Limitation: Finding suitable recipients and coordinating timely collection can be challenging.

2.SURPLUS FOOD MARKETPLACES:

Example: Too Good To Go

What they do: Help businesses sell surplus food at discounted prices.

Limitation: Selling surplus after preparation does not necessarily prevent overproduction.

3. MANUAL TRACKING & COORDINATION:

Examples: Spreadsheets, phone calls, WhatsApp groups

What they do: Track leftover food and contact potential recipients.

Limitation: Fragmented information and reactive coordination can delay redistribution and make recurring waste patterns harder to identify.

THE GAP — SMARTFOOD:

From reactive surplus handling to predictive food-waste management

1.PREDICT: Forecast demand and identify potential surplus early.
2.PREVENT: Reduce avoidable overproduction.
3.MATCH: Find suitable recipients based on need, capacity and distance.
4.COORDINATE: Prioritize timely pickup and track redistribution.
5.LEARN: Use historical data to improve future decisions

KEY TAKEAWAY

Existing approaches address parts of the problem. SmartFood aims to integrate prediction, prevention and time-sensitive redistribution into one intelligent workflow to reduce avoidable surplus and help more edible food reach people who need it.

PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem

- **Hardware Requirements**
- **Laptops/Computers** – for AI model development and platform management
- **Smartphones** – for businesses and recipient organizations to report and receive surplus alerts
- **Cloud infrastructure** – for storing data and running the platform
- **Software Requirements**

Frontend (User Interface):

- - React.js – for building the web application.

Backend (Server & Logic):

- - Python + FastAPI – for backend services and AI integration.

AI & Machine Learning:

- - Python, Pandas and Scikit-learn – for demand forecasting and surplus prediction.

Database:

- - PostgreSQL – for storing food listings, recipient details, demand patterns and pickup information.

Maps & Route Optimization:

- - OpenStreetMap – for location and map data.
- - OSRM – for calculating efficient pickup routes.

Development & Deployment:

- - Visual Studio Code – for writing and managing code.
- - GitHub – for version control and team collaboration.
- - Streamlit or a cloud hosting platform – for demonstrations and deployment, if required.

- **Other resources (Including Human Resources)**

- **Food providers**: Restaurants, hotels, supermarkets, cafeterias and event organizers
- **Recipients:** NGOs, shelters, food banks and community organizations
- **Pickup network:** Delivery partners or volunteers
- **Data:** Historical demand, production and surplus data
- **Food-safety guidelines** for responsible redistribution

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INITIAL DIRECTION & EXPECTED OUTCOME

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What Is Your Expected Outcome?

Initial Direction:

- Develop an AI-powered platform to predict potential food surplus using historical sales and demand data.
- Help businesses reduce unnecessary food preparation through demand forecasting.
- Match unavoidable edible surplus with suitable nearby NGOs and food banks.
- Coordinate timely pickups through notifications and backup recipient matching.

Expected Outcomes:

- Reduce avoidable food overproduction and food waste.
- Improve coordination between food donors and recipient organizations.
- Increase the quantity of suitable surplus food successfully redistributed.
- Help businesses reduce losses associated with excess food preparation.
- Track food prevented from being wasted and food successfully redistributed.

Success Indicators:

- Accuracy of surplus predictions.
- Quantity of avoidable surplus prevented.
- Quantity of food successfully redistributed.
- Time taken to match donors with recipients.
- Successful pickup rate.